

Face Mask Detection with Live Alert System

Elevate Labs — AI/ML Virtual Internship Project Report

1. Introduction

The COVID-19 pandemic made face masks a critical tool for reducing the spread of airborne illness in public and semi-public spaces. Manually monitoring mask compliance at entrances, offices, or crowded areas is labor-intensive and error-prone. This project implements an automated, real-time face mask detection system that uses a webcam feed to identify whether individuals are wearing a mask and raises a live alert when non-compliance is detected, offering a lightweight, deployable alternative to manual monitoring.

2. Abstract

The system combines classical computer vision with deep learning. OpenCV's Haar Cascade classifier localizes faces in each video frame, and each detected face is passed to a MobileNetV2-based convolutional neural network, fine-tuned via transfer learning on a labeled dataset of masked and unmasked faces, to classify it as 'Mask' or 'No Mask'. Detections are rendered as color-coded bounding boxes with confidence scores, and a persistent on-screen alert is triggered when a face without a mask is detected for more than a short time threshold. The model achieved strong classification performance on a held-out test split, and the full pipeline runs in real time on a standard laptop webcam.

3. Tools Used

- Python 3.9+ — core implementation language
- OpenCV — face detection (Haar Cascades) and webcam video I/O
- TensorFlow / Keras — MobileNetV2 CNN, transfer learning, model training
- scikit-learn — train/test split and classification metrics (precision, recall, F1)
- Matplotlib — training loss/accuracy curve visualization
- Flask (optional) — web interface for single-image mask detection

4. Steps Involved in Building the Project

- 1 Collected a labeled dataset of masked and unmasked face images (Kaggle) and organized it into with_mask / without_mask folders.
- 2 Preprocessed images: resized to 224x224, converted to arrays, normalized using MobileNetV2's preprocessing function.
- 3 Built a transfer-learning classifier: a frozen MobileNetV2 base with a custom head (average pooling, dense, dropout, softmax).
- 4 Trained the model with data augmentation (rotation, zoom, shift, flip) to improve generalization, and evaluated it with a classification report.
- 5 Integrated the trained model with OpenCV's Haar Cascade face detector to process a live webcam stream frame by frame.
- 6 Added bounding-box visualization with color coding (green = mask, red = no mask) and confidence percentages.
- 7 Implemented alert logic that triggers an on-screen warning and console log when a no-mask face persists beyond a set duration.
- 8 Optionally deployed a Flask web app allowing users to upload a static image and receive an annotated result.

5. Conclusion

This project demonstrates an end-to-end computer vision pipeline, from dataset preparation and transfer-learning-based model training to real-time inference and alerting. The resulting system reliably distinguishes masked from unmasked faces in live video and could be extended with a more robust deep-learning face detector, automated notification (email/SMS), and event logging for deployment in

real-world settings such as building entrances or workplaces.